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Tap vs Bottled: Assessing PFAS Levels and Exposure Risks in Singapore's Drinking Water

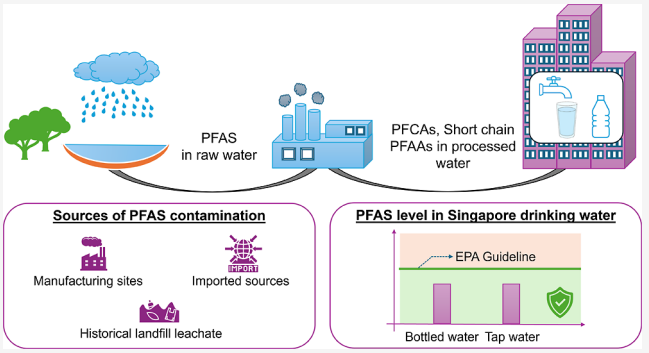
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ABSTRACT: Globally, drinking water choices typically involve tap or bottled water, which is influenced by perceptions of quality and safety. Raw water sources are generally considered unsafe due to potential diseases and contaminants. The presence of per- and polyfluoroalkyl substances (PFASs) in drinking water has raised health concerns due to their persistence, toxicity, and potential long-term health impacts. In Singapore, tap water is the primary household drinking source, with bottled water as a supplement. Analysis of both revealed low PFAS levels, indicating minimal health risks. Detected PFAS concentrations ranged from 0.30 to 15.17 ng/L in tap water and from 0.05 to 2.45 ng/L in bottled water, well below the US Environmental Protection Agency limits. The PFAS profile primarily consisted of C4−C9 perfluorocarboxylic acids (PFCAs) and perfluorosulfonic acids, with major contributions from PFCA and short-chain perfluoroalkyl acids (PFAAs) containing 5 or fewer CF2 groups. This study emphasizes the need for advanced technologies to remove short-chain PFASs from drinking water and proposes future research directions to protect public health.



KEYWORDS: *perfluoroalkyl substances, exposure assessment, risk assessment, tap water, bottled water, public health*

1. INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) make up a class of synthetic chemicals widely used in industrial and consumer applications due to their unique properties, including chemical stability and ability to reduce aqueous surface tension. These make them desirable for use in aqueous film forming foams, cookware, furniture, food-contact papers, cosmetics, cleaning products, stain-resistant fabrics, and many other applications.[1] However, their resistance to degradation and persistence in the environment have earned them the moniker "forever chemicals". Once released, PFASs remain in the environment for extended periods and are challenging to remove, leading to widespread contamination of air,[2] soil,[3] water,[4−6] and human exposure. They enter the human body through air and dust inhalation, dermal contact, and dietary intake. Dietary exposure is identified as the major pathway when occupational exposure is absence.[7] Food is recognized as the prominent contributing factors to dietary exposure for the majority of the study and participants, while water could not be neglected as we drink and cook food using water commonly.

Water contamination of PFASs occurs through multiple pathways, including industrial discharge, atmospheric deposition, and groundwater infiltration. Sources of contamination often include manufacturing sites, areas of heavy PFAS-containing commodity usage, and landfill leachates.[8]

Studies have linked PFAS exposure to developmental effects during early life stages, immune system disorders, obesity, kidney disorders, increased cancer incidence, and endocrine disruptions.[9−12] These health risks underscore the necessity of monitoring and regulating PFAS in drinking water to protect public health.

The Stockholm Convention has listed PFASs as persistent organic pollutants, calling for global action to tighten the legislation on PFASs in water. In the United States, the EPA has established maximum contaminant level guidelines of 4 ng/L for PFOA and PFOS and 10 ng/L for PFHxS, PFNA, and Gen X individually. A hazard index of 1 is the maximum allowable content when there is a mixture with two or more of PFHxS, PFNA, PFBS, and Gen X in tap water.[13] This is also applicable to bottled water, as long as no disparity of enforceable level is proposed by the U.S. Food and Drug Administration (FDA).[14] The European Union (EU) mandates a limit of 100 ng/L for the sum of 20 PFAS and 500 ng/

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L for total PFAS in drinking water, according to Directive (EU) 2020/2184. The World Health Organization (WHO) proposes provisional guidelines for drinking water quality, with an individual provisional guideline value of 100 ng/L for PFOS and PFOA and a total value of 500 ng/L for all PFASs. In Australia and New Zealand, PFOA is allowed at 560 ng/L, and the sum of PFOS and PFHxS is allowed at 70 ng/L for tap water. Bottled water in these countries adheres to the WHO guideline. In Canada, the maximum limit for PFOA in tap water is 200 ng/L, while that for PFOS is limited at 600 ng/L. Additionally, the sum of PFOS and PFOA concentrations in drinking water divided by their respective limits should not exceed 1. Regional variation in the thresholds is observed, reflecting differing approaches to PFAS regulation worldwide and likely differences in the methodologies employed to quantify them.

Despite these regulatory efforts by the environment and food safety authorities of the developed countries, the knowledge gap on PFAS occurrence and risk profile persists, particularly in Southeast Asia, where PFAS contamination in drinking water remains underexplored. Countries such as Thailand,[15] Philippines,[16] and Malaysia[17] have documented PFAS footprints, yet comprehensive surveillance is scarce. In Singapore, no large-scale studies of PFASs in drinking water have been conducted, highlighting the need for targeted investigations.

As a small nation with limited natural resources, Singapore's water sources are highly diversified, ranging from imported water to reclaimed rainwater, desalinated seawater, and recycled wastewater.[18] The complexity of raw water sources necessitates stringent water quality control (QC) to ensure drinking water safety. The raw water must be treated before distribution to households to ensure that the quality of drinking water meets the standards set forth in the Environmental Public Health (Water Suitable for Drinking) (No. 2) Regulations 2019, which are based on WHO guidelines. Singapore has participated in the Stockholm Convention since May 2001 and practices stringent water quality monitoring. In this study, we monitored the occurrence level of PFAS in tap and bottled water in Singapore to examine the associated risks of extensive daily water consumption from cooking and drinking. By analyzing daily water intake and the occurrence of PFAS, we establish a risk profile of local consumers considering their daily water intake and PFAS occurrence levels. Surprisingly, we find a consistent PFAS footprint in tap water across all regions of Singapore, despite the diverse sources of water supply. Our analysis indicated that local consumers are not at high risk from PFAS exposure through drinking water, and we proposed directions for future research on PFASs.

2. MATERIAL AND METHODS

2.1. Scope of Targeted PFAS. In this study, we focused on 14 PFASs (see Table S1), specifically targeting C4−C14 perfluoroalkyl acids (PFAAs) and GenX. These compounds are commonly regulated and well-studied for their toxicity, indicating that chronic exposure to PFASs above advisory limits may pose health risks.[13] Specifically, existing toxicology studies are limited and cannot adequately assess the carcinogenic potential of PFBA exposure; therefore, it is not monitored in this study.[19] PFAAs with ultrashort (<3 carbon) and long (>15 carbon) carbon chains are not monitored in this study as they are reported to have poor SPE recovery.[20,21] Separately, PFAS precursors such as 6:2 fluorotelomer sulfonic acid (6:2 FTSA), 8:2 fluorotelomer sulfonic acid (8:2 FTSA), and methyl perfluorooctane sulfonamido are not monitored due to their potential to degrade and transform into other types of PFASs in the environment.[22] Commercial availability of reference materials for these PFASs is also limited.

2.2. Standards and Chemicals. Native and isotopic mass-labeled internal standards, PFPeA, PFHxA, PFHpA, PFOA, PFNA, PFDA, PFDoA, PFTriA, PFTeDA, [$^{13}C_5$] PFPeA, [$^{13}C_9$] PFDA, [$^{13}C_2$] PFTeDA, [$^{13}C_3$] PFBS, [$^{18}O_2$] PFHxS, and [$^{13}C_3$] HFPO−DA, were purchased from Wellington Laboratories (Canada). PFBS, PFHxS, PFOS, HFPO−DA, PFUnA, [$^{13}C_6$] PFHxA, [$^{13}C_8$] PFOA, [$^{13}C_4$] PFOS, [$^{13}C_9$] PFNA, [$^{13}C_9$] PFUnA, and [$^{13}C_{12}$] PFDoA were purchased from Cambridge Isotope Laboratories (United States). The full list of reference materials is summarized in Table S1. Liquid chromatography−mass spectrometry (LC−MS) grade methanol (certified to be PFAS-free) and formic acid were purchased from Fisher Scientific (United States). ACS grade ammonia hydroxide was purchased from Merck (Germany). MS grade ammonium acetate was purchased from Merck (United States). All chemicals were used without further purification. Polypropylene disposable pipets and containers demonstrated to be free of interference were used to avoid cross-contamination. Glassware was washed with a laboratory-grade detergent, multiple tap water rinses, followed by a Millipore water rinse, and finally methanol to minimize PFAS interferences.

2.3. Sampling Regime and Sources. Singapore residents consume both tap and bottled water, with choices driven by age, economic factors, and health considerations.[23−25] Our study analyzed 164 water samples, comprising 87 tap and 77 bottled water. Tap water was collected from local households between June and July 2023, while bottled water samples were collected from local commercial sources between June and December 2023. The tap water sampling procedure was conducted systematically to ensure the sample integrity and prevent contamination. Samples were collected by personnel wearing clean nitrile gloves, with tap water being flushed for 5 min before collection into a 250 mL polypropylene bottle. Each bottle was filled to 90% capacity to prevent sample overflow and to avoid potential contamination from the outer surface of the containers. The samples were delivered and stored at 20 °C for up to 5 days before extraction to preserve the pristine PFAS content. Sample storage conditions deviated from EPA 537.1[26] temperature guidelines was based on previous research demonstrating PFAS stability in water at room temperature. A study of 29 PFAS compounds showed minimal concentration changes in bottled water under ambient conditions.[27] Considering Singapore's high-quality tap water treatment standards, we anticipated similar stability behavior. This approach was supported by our QC measures, including a 5 day storage limit and recovery studies (75−125% recovery rates). Temperature-sensitive PFAS compounds were excluded from the analysis to ensure data reliability. Future studies will implement the EPA's recommended storage temperature of below 6 °C, particularly for temperature-sensitive compounds. Carbonated, oxygenated, and hydrogenated water underwent pretreatment via ultrasonication at room temperature for 30 min prior to extraction.

Tap water was collected from 42 locations across all regions of Singapore,[28] specifically 24 samples from the North (n = 24), 12 from the East (n = 12), 6 from the Northeast (n = 6),

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27 from the West (n = 27), and 18 from Central (n = 18). The sampling locations were labeled in Figure S1 and were representative for the tap water consumption in the residential areas. The water samples were stored in PFAS-free polypropylene bottles, which have been tested with migration studies and shown to leach negligible amounts of PFASs under the experimental conditions.

To have a good representative of bottled water on the Singapore retail market, the total import volume from 2019 to July 2023 (Figure S2) was analyzed to guide the sampling regime. Malaysia imports dominated the total import, at 89.1%. Products from other countries and regions, such as Europe, Asia, Australia, and North America, contributed significantly smaller volumes of less than 5%. Bottled water was purchased from local supermarket and grocery stores, targeting the top 5 importing countries, namely, Malaysia, France, Indonesia, Italy, and Taiwan. Details of the samples are tabulated in Table S2.

2.4. Sample Preparation. All bench surfaces were precleaned with LC−MS grade MeOH and allowed to dry before sample preparation, with strict protocols prohibiting contact with PFAS-containing materials such as tissue papers and aluminum foil. The internal standard addition method was deployed for PFAS quantification in this study. In brief, 50 μL of a 20 ng mL^{-1} surrogate isotopic labeled internal standard mixture was added to 100 mL of sample. 5 μL of formic acid was added to acidify the mixture and thoroughly mixed prior to solid-phase extraction with an Oasis mixed-mode weak anion-exchange (WAX) reversed-phase cartridge (3 cc, 60 mg sorbent, 30 μm particle size; Waters Corporation, Unites States). The Oasis SPE cartridge is chosen because it can achieve satisfactory recovery for the compounds of interest. Solvent reservoirs and vacuum extraction chamber were rinsed with methanol and 18.2 MΩ cm ultrapure water (Millipore) twice to remove background residue PFAS contamination. The cartridges were preconditioned and activated with 4 mL of methanol and 4 mL of Millipore water sequentially. The sample was then loaded to SPE at a speed of 1−2 drops per second. The cartridges were rinsed with Millipore water, 3% formic acid in water, and methanol in sequence after loading. The cartridges were then dried by vacuum for 10 min. The elution was performed by adding 4 mL of 1% ammonium hydroxide in methanol. The extract was evaporated to dryness with nitrogen (Caliper LifeSciences TurboVap LV, MA, USA) and reconstituted in 200 μL of methanol/water (4:1, v/v). The solution was centrifuged (Hermle Z383 K, BW, Germany) and filtered with an inorganic membrane filter (Whatman, 0.1 μm, 10 mm) before being transferred to a LC sample vial for analysis. Extracts were analyzed within 14 days from the day of sampling.

2.5. Instrumental Analysis. PFAS analysis was performed on a LC−MS/MS instrument, composed of an Agilent Technologies 1290 Infinity II UHPLC system interfaced with Sciex Triple Quad 6500+ MS. In order to minimize instrumental PFAS contamination, all the fluoropolymer tubing was replaced with polyether ether ketone or stainless-steel tubing. An InfinityLab PFC delay column (30 × 4.6 mm) was installed downstream of the LC pumps to segregate the background signal from the sample signal. Chromatographic separation of 35 μL of injection was achieved on a reverse phase InfinityLab Poroshell 120 EC-C18 analytical column (100 × 2.1 mm, 2.7 μm). The flow is a multistep aqueous−organic (1.25 mM ammonia acetate−methanol) gradient lasting for 25 min at 0.3 mL/min (Table S3). Mass spectra were obtained in the negative electrospray ionization mode. Compounds were detected in the multiple reaction monitoring (MRM) mode with reference to the quantifier and qualifier MRM transition. The chromatogram of spiked standard and internal standard with eluting order is shown in Figure S3. The optimized parameters of MS, individual analytes, and isotopically labeled internal standards are summarized in Tables S4−S6.

2.6. Method Validation and Data Analysis. Method validation was performed in reference to EPA methods and revalidated with in-house available materials.[26,29] Calibration curves were constructed within the range of 0.5−10.0 μg/L. External calibration and matrix-matched calibration were conducted for method validation. The calibration was conducted by using both external and matrix-matched calibration approaches. For external calibration, we constructed a six-point calibration curve for analysis. Matrix-matched calibration was performed to evaluate potential matrix effects by adding standard and surrogate isotope-labeled internal standards to ultrapure water, followed by extraction using the sample preparation procedure detailed in Section 2.4. This dual calibration approach enabled accurate assessment of method performance and quantification of matrix-induced variations in the analyte response. Good linearity within the calibration range of external and matrix-matched calibration was achieved, with a correlation factor (R^2) greater than 0.99 (Table S7). The spike recovery of external calibration matched the linearity of matrix-matched calibration, confirming that external calibration is suitable for routine analysis. The estimated method detection limits (MDLs) were within the range from 0.06 to 0.11 ng/L (Table S8), as a result of 3.3 times of the standard deviation of linear regression divided by the gradient of the external calibration curve. PFASs were quantified and reported in their acidic form, calculated with the relative response of analytes to their respective internal standards. Recovery of spikes with external calibration ranged from 82 to 114% and recovery of spikes with a matrix-matched calibration curve ranged from 82 to 116%. Matrix spikes at 2, 3, and 5 ppb bear a minimum recovery at 95.5% and a maximum recovery at 113.1%, all within ±15%. The relative standard deviation of all compounds were all <20% (Table S7). Procedural and sampling blanks were employed to assess potential sampling, laboratory contamination, and interferences for every run. Samples were flagged for contamination when blank values exceeded one-third of the reporting limit. Affected samples with acceptable procedural contamination levels were re-extracted and reanalyzed upon investigation of the contamination sources and thorough cleaning with methanol, while samples with heavy procedural and sampling contamination were excluded from the study results. Matrix spikes in the middle range of assay were included in each batch as QCs, and their recoveries were between 90 and 110%. The standard solution was injected after every 9 samples to monitor the instrument performance. The accuracy of the analysis of the QC was within 90−110%.

Mass compositional analysis was performed to evaluate the contamination pattern in chain length, functional group, and compound abundancy. The relative mass composition is calculated as the percentage of the total mass of a specific PFAS compound against the total PFAS mass.

$$\sum \text{Compound} \geq \text{MDL} / \sum_{14} \text{PFAS}$$

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Table 1. Summary of PFAS Detections in Bottled (n = 77) and Tap Water (n = 87)[f]

	bottled water				tap water			
analyte	detection (Frequency)	mean[a]	median[a] [Min[b], Max]	relative mass abundance[c] (%)	detection (frequency)	mean[a]	median[a] [Min[b], Max]	relative mass abundance[c] (%)
$\sum^{PFCA}$	44 (57%)	0.25	0.17 [0.05, 0.96]	73.60	87 (100%)	0.79	0.49 [0.06, 4.23]	73.10
PFPeA[d]	25 (32%)	0.31	0.23 [0.08, 0.88]	30.60	67 (77%)	1.56	1.35 [0.11, 3.51]	21.60
PFHxA[d]	16 (21%)	0.32	0.26 [0.06, 0.94]	20.20	87 (100%)	1.36	1.00 [0.21, 4.63]	24.40
PFHpA[e]	12 (16%)	0.11	0.10 [0.07, 0.17]	5.00	84 (97%)	0.51	0.25 [0.07, 1.79]	8.70
PFOA[e]	6 (8%)	0.33	0.34 [0.07, 0.55]	7.80	82 (94%)	0.59	0.41 [0.14, 1.78]	9.90
PFNA[e]	13 (17%)	0.16	0.11 [0.05, 0.42]	8.30	85 (98%)	0.39	0.30 [0.08, 1.25]	6.70
PFDA[e]	1 (1%)	0.07		0.30	34 (39%)	0.16	0.12 [0.07, 0.42]	1.10
PFUnA[e]	1 (1%)	0.11		0.40	22 (25%)	0.11	0.10 [0.06, 0.21]	0.50
PFDoA[e]	1 (1%)	0.16		0.60	1 (1%)	0.06		0.01
PFTriA[e]	1 (1%)	0.08		0.30	1 (1%)	0.23		0.05
PFTeDA[e]	ND	ND			1 (1%)	0.19		0.04
∑PFSA	12 (16%)	0.24	0.10 [0.06, 0.85]	14.40	80 (92%)	0.68	0.55 [0.06, 2.23]	26.10
PFBS[d]	7 (9%)	0.37	0.13 [0.06, 0.83]	10.20	73 (84%)	1.14	1.04 [0.07, 1.99]	17.20
PFHxS[e]	5 (6%)	0.1	0.09 [0.08, 0.14]	1.90	55 (63%)	0.29	0.17 [0.06, 0.97]	3.30
PFOS[e]	4 (5%)	0.16	0.10 [0.07, 0.36]	2.40	57 (66%)	0.49	0.26 [0.06, 2.18]	5.70
∑PFECA	14 (18%)	0.22	0.19 [0.08, 0.39]	11.90	8 (9%)	0.55	0.27 [0.14, 1.53]	0.70
GenX	14 (18%)	0.22	0.19 [0.08, 0.39]	11.90	8 (9%)	0.44	0.28 [0.14, 1.50]	0.70
Short-chain PFAA[d]	52 (68%)	0.32	0.24 [0.06, 0.94]	72.90	87 (100%)	1.35	1.21 [0.07, 4.13]	63.80
Long-chain PFAA[e]	40 (52%)	0.16	0.11 [0.05, 0.55]	27.10	87 (100%)	0.41	0.26 [0.06, 2.18]	36.10
$\sum_{14}$PFAS	52 (68%)	0.33	0.20 [0.05, 2.45]	100	87 (100%)	5.58	4.28 [0.30, 15.17]	100

[a]Calculated mean and median are based on detections above MDL. [b]Minimum results above MDL are reported. [c]Fraction of total PFAS mass detected in across all samples = $\sum_{n=1}^{87} \frac{[\text{compound} \geq \text{MDL}]}{[\sum \text{PFAS}]}$ [d]Short-chain PFAA = C3−C6 PFCA + C3−C5 PFSA. [e]Long-chain PFAA = C7−C10 PFCA + C6−C8 PFSA. [f]Concentration in ng/L.

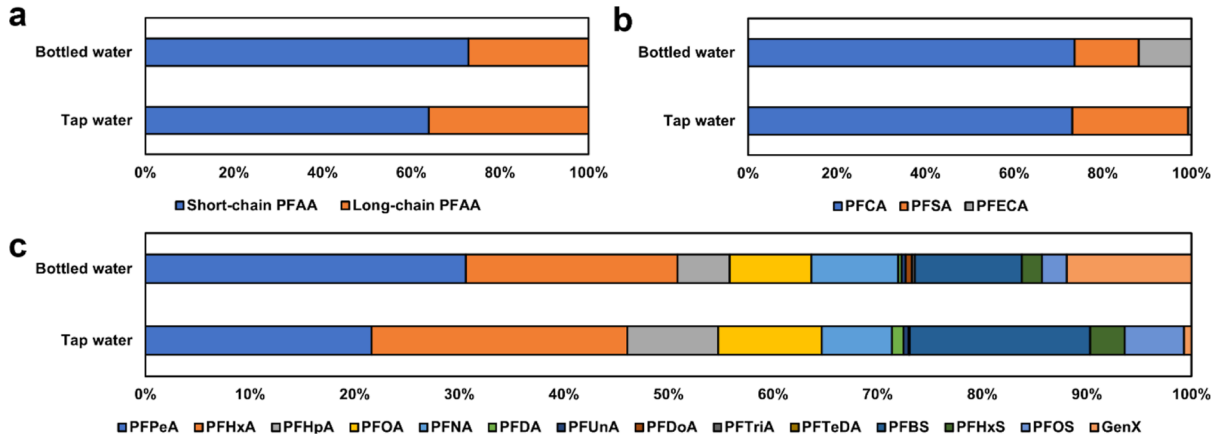


Figure 1. Relative mass composition profile of PFASs. (a) Short-chain PFAA and long-chain PFAA composition; (b) PFCA, PFSA, and PFECA composition; (c) compound-specific composition in bottled and tap water.

Human risk exposure was evaluated based on the comparison of measured concentration with that of the established tolerable weekly intake (TWI) by the European Food Safety Authority (EFSA), 4.4 ng/kg bw/week for the total PFOS, PFOA, PFNA, and PFHxS content. The daily intake (DI) of PFASs from water is calculated according to the following formula:

$$DI = C \times Q/Bw$$

Here, C represents the concentration of PFASs in water (ng/L). Q is the daily drinking water consumption (L/day). Bw is the average human body weight (kg). The food consumption data from the 24 h dietary recall interview conducted in Singapore Food Agency's Total Diet Study (TDS) in 2021 was used to conduct the exposure assessment.[30] The survey was conducted with participants aged 15−92 from the Singapore general population.

Kruskal−Wallis (KW) ranked sum tests were used to compare individual and total PFAS concentrations on the basis of categorical groupings. Significance was set at a threshold of $p < 0.05$. Mann−Whitney−Wilcoxon (MWW) test were used to compare individual and summed PFAS concentration of bottled water. Nondetects below MDLs were included and assigned the average lowest ranking.

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3. RESULTS AND DISCUSSION

3.1. PFAS Contamination Pattern in Water. Fourteen PFAS compounds, including short-chain to long-chain PFAAs, perfluorocarboxylic acids (PFCAs), perfluorosulfonic acids (PFSAs), as well as perfluoroether carboxylic acid (PFECA), were monitored in this study (Table 1). All 14 compounds were detected in at least one water sample above their respective MDLs. In total, 68% of bottled water (52 out of 77) and all tap water (87 out of 87) were detected with at least one type of PFAS, with PFAA comprising 4−9 carbon chain lengths being frequently detected (Figure S4).

Chain length analysis revealed a predominant contamination of short-chain PFAAs in both tap water and bottled water. Short-chain PFAAs accounted for a relative mass contribution of 72.9% in bottled and 63.8% in tap water (Figure 1a and Table 1). The most frequently detected short-chain PFAAs in tap water were PFHxA (100%), PFBS (84%), and PFPeA (77%). Similarly in bottled water, PFPeA (32%) and PFHxA (21%) were also among the most frequently detected PFASs. In contrast, long-chain PFAAs had a lower relative mass abundance, accounting for less than 36.1% as compared to short-chain PFAAs. Their detection frequency in tap water ranged from 66% to 97%, including compounds such as PFHpA, PFOA, PFOS, PFNA, and PFUnA. However, in bottled water, these long-chain PFAAs were detected less frequently, with detection rate ranging from 5 to 17%. Other long-chain PFAAs, such as PFDA, PFDoA, PFTriA, and PFTeDA, were found negligible, with detection frequencies less than 1%. These long-chain PFAAs tend to adsorb favorably to sediments, making them less soluble and mobile in water.[31] Despite global phase out and regulatory measures undergone, the detection of PFOA and PFOS in the tested tap and bottled water samples indicates that legacy PFAS contamination continues to affect the drinking water systems. Ongoing research and technology innovation for legacy PFAS cleanup remain essential. Additionally, PFNA and other volatile PFAS compounds such as fluorotelomer alcohols can contaminate natural water sources through atmospheric precipitation, posing a risk to our drinking water supply.[32] The presence of PFNA, PFPeA, PFOA, PFHxA, and PFHpA in tap water may be attributed as degradation products of volatile compounds, such as fluorotelomer alcohols.[33]

Functional group profiling indicated that PFCAs were the predominant contaminants (Figure 1b), accounting for over half of the total contamination, specifically, 73.1% in tap water and 73.6% in bottled water. In contrast, PFSAs contributed a lower relative mass, ranging from 14.4% to 26.1%. A detailed analysis of compound profiles revealed a similar relative mass composition in both bottled and tap water, with PFPeA, PFHxA, and PFBS being the most prevalent, accounting for 14.1% to 30.6% (Figure 1c).

Our findings on the higher mass contribution of short-chain PFAAs and PFCAs align with global observation. This trend is likely due to the limitation of conventional water purification technologies in effectively removing short-chain PFAAs. The existing purification processes primarily rely on granular activated carbon (GAC) and anion exchange resin (AIX), which work with hydrophobic and ionic interactions, respectively.[4,34] Long-chain PFAAs, characterized by their extended fluorinated backbones, can be effectively removed by GAC through a hydrophobic interaction. However, the removal of short-chain PFAAs is more challenging because of their reduced hydrophobicity. Instead, they rely on the ionic interaction with AIX for removal, but their removal efficacy is often compromised by matrix interference and competition with other ions for limited adsorption sites, resulting into poor removal efficacy.[35] Moreover, the acidic headgroups of PFCAs create weaker ionic interaction with AIX compared to PFSAs, leading to PFCAs' inferior removal efficiency.[34] This differential interaction helps to explain the predominance of PFCAs and short-chain PFAAs in treated water and highlights the urgent need for advanced purification technologies to be specifically developed for the targeted removal of these PFAS.

Analysis of PFAS levels in tap water using the KW ranked-sum test revealed no significant differences in the concentrations of aggregated short-chain, long-chain, and total PFAS across the country, despite diverse sources of water supply (Table S9). This negligible variation indicates that upstream water QC measures implemented in Singapore are consistently effective, particularly due to the advanced treatment technologies currently employed. Singapore's diverse raw water sources, including rainwater, seawater, and wastewater, have distinct PFAS contamination profiles.[36−38] However, our study demonstrates that comprehensive water treatment processes effectively mitigate the variations, producing good quality processed water with a low PFAS content suitable for household consumption. The water treatment process in Singapore comprises several stages, including chemical treatment, filtration, and disinfection.[39] Specifically, it involves coagulation, sedimentation, and filtration utilizing sand filters or membrane systems for sediment removal, subsequently followed by chlorine and ozone disinfection and passage through biologically activated carbon filters. The treated water is then stored in tanks and distributed via service reservoirs. This robust approach for water treatment and management has resulted in uniformly high-quality tap water throughout Singapore, with minimal regional variations in the PFAS concentrations. We recognize several important factors that warrant further investigation, including potential seasonal variations in PFAS levels between dry and wet periods and the impact of land runoff on PFAS concentrations in water supply. To address these limitations in future studies, we propose implementing year-round sampling to capture seasonal variations, investigating the relationship between weather patterns and PFAS levels, and conducting detailed analysis of source water variations.

In the analysis of bottled water, the MWW test revealed no statistically significant differences in aggregated PFAS concentrations between drinking water and mineral water (Table S10). Furthermore, the KW test showed no significant variations among bottled water with various packaging materials, including PET, rPET, and glass (Table S11). Notably, recycled PET and virgin PET exhibited comparable levels of PFAS contamination, suggesting that neither the packaging manufacturing process nor subsequent migration introduces additional PFAS contamination into the water. However, it is important to note that natural variations in water sources for bottled water production may obscure the potential impact of packaging materials on PFAS levels. To address this limitation, future research should conduct separate migration studies using intact unadulterated materials for a more accurate comparison.

3.2. Occurrence Levels and Benchmarking against Regulatory Limits. The total PFAS occurrence levels differed in different water categories, with bottled waters being the least

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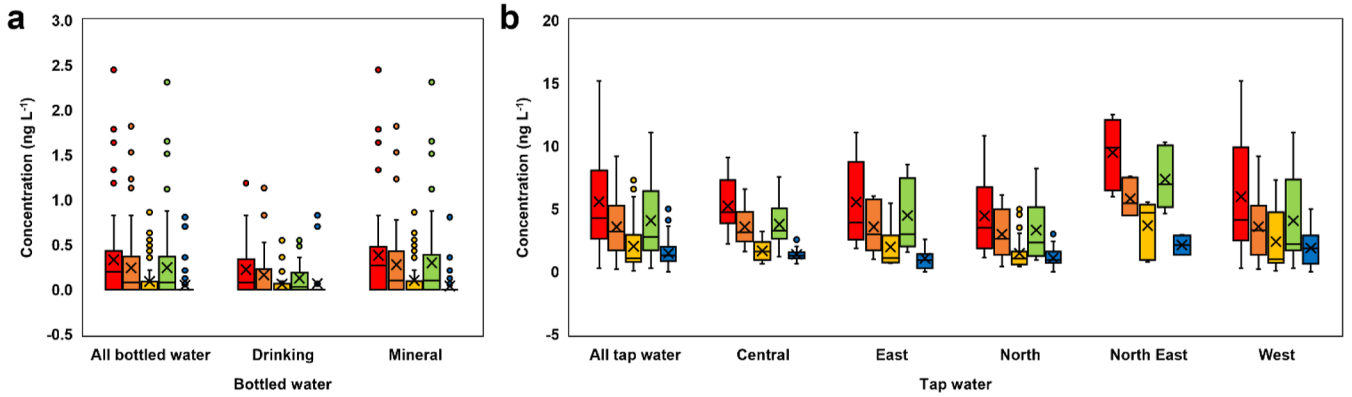


Figure 2. PFAS occurrence level distribution in (a) all bottled water, drinking water, and mineral water and (b) all tap water, tap water collected from the Central, East, North, North East, and West regions of Singapore. Red box—$\sum_{14}$PFAS. Orange box—short-chain PFAS. Yellow box—long-chain PFAS. Green box—PFCAs. Blue box—PFSAs.

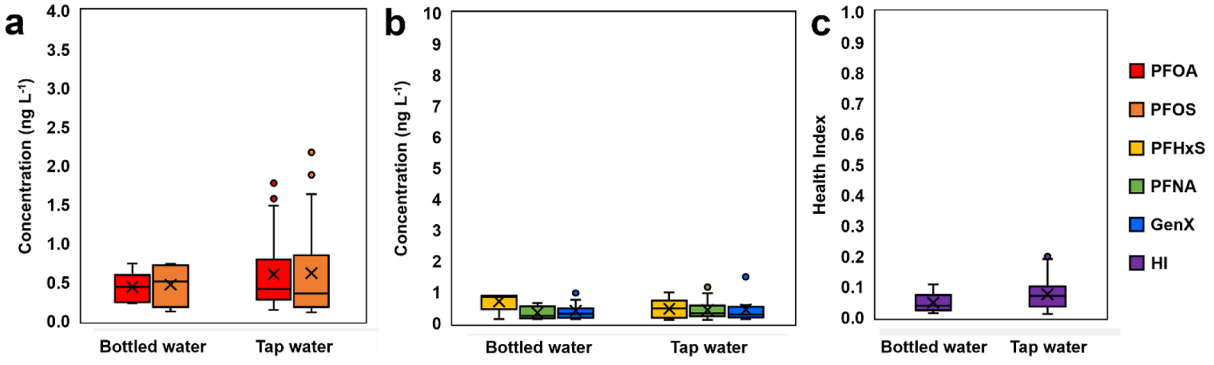


Figure 3. Occurrence of PFASs as compared to EPA guidelines. (a) Individual PFOS, PFOA shall not exceed 4 ng/L. (b) Individual PFHxS, PFNA, Gen X shall not exceed 10 ng/L. (c) Health index of combined PFASs shall not exceed 1 following this equation. Box plot illustrates the occurrence level is far below the stipulated limits. Red box—PFOA. Orange box—PFOS. Yellow box—PFHxS. Green box—PFNA. Blue box—GenX. Purple box—Health Index.

contaminated categories. The sum of 14 PFAS concentrations, $\sum_{14}$PFAS (Figure 2a,b, red bar), ranged from 0.05 to 2.45 ng/L in bottled water and 0.30−15.17 ng/L in tap water. The mean occurrence level in bottled water was 0.34 ng/L (Table 1), while 5.79 ng/L was estimated in tap water (Table 1). The PFAS contamination level in the bottled water is about 20 times lower in mean concentration than that in tap water.

In bottled water, the mean $\sum_{14}$PFAS concentrations in drinking and mineral water were 0.22 and 0.38 ng/L, respectively. The concentration ranges were 0.06−1.19 ng/L for drinking water and 0.05−2.45 ng/L for mineral water (Figure 2a). In our study, the PFAS footprint in mineral water exceeded that of drinking water only marginally. This finding contrasts with the detection of PFAS in U.S. bottled water, where mineral water was found to be more contaminated, with the highest total PFAS concentrations reaching 18.87 ng/L.[6] Our mineral water samples were much less contaminated compared to the most contaminated mineral water sampled in the United States. This suggests that the bottling sources of the mineral water in our study may have remained largely unaffected by heavy PFAS contamination.

The mean concentrations of $\sum_{14}$PFAS in tap water across different regions of Singapore were 5.23 ng/L (Central), 5.54 ng/L (East), 9.48 ng/L (North), 4.44 ng/L (North East), and 5.97 ng/L (West), respectively. The concentrations ranged from 2.22 to 9.10, 1.85 to 11.07, 5.99 to 12.51, 1.13 to 10.83, and 0.30 to 15.17 ng/L for these regions (Figure 2b).

The analysis of total PFAS concentrations in Singapore's tap water indicated relatively low levels of contamination when compared to other countries[40−45] (Table S12). Notably, the maximum occurrence levels of PFOA, PFOS, PFNA, and PFHxS, being the most prominent PFASs found in human blood serum,[7] were among the lowest observed (Table S13). These levels are approximately 2-order-of-magnitude lower than those reported in Malaysia and South Korea.

These findings indicate the absence of significant point sources or contamination hotspots within the country. The observed low PFAS levels can be partly attributed to the limited manufacturing activities related to PFASs in Singapore, which inherently reduce the risk of water system contamination. This situation is similar to the findings of tap water studies conducted in Turkey,[46] where comparable low levels of PFAS contamination have been reported.

The relatively low PFAS concentrations in Singapore's tap water underscore the effectiveness of the country's water management strategies and pollution prevention measures. This outcome is particularly noteworthy in light of global concerns about PFAS contamination of water resources. The data not only reflect positively on Singapore's water quality but also provide valuable insights for comparative studies in urban water management across different geographical contexts.

We benchmark the PFAS occurrence level of bottled and tap water to the maximum allowable limits established by the EPA. The EPA sets individual limit for PFOA and PFOS at 4 ng/L,

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and for PFNA, PFHxS, and GenX at 10 ng/L, while the hazard index shall not exceed 1. In this study, the PFAS occurrence levels detected in both tap and bottled water samples are consistently less than half of the stipulated limits (Figure 3a–c), demonstrating that the water quality fulfills the requirements of the EPA and is safe for consumption.

3.3. Dietary Exposure and Risk Assessment. The toxicological profiles of PFOS, PFOA, PFNA, and PFHxS are well-established, with Tolerable Weekly Intake (TWI) values of 4.4 ng/kg bw/week proposed by the EFSA.[7] This value is derived to protect infants against decreased vaccination response when receiving breastmilk from adult mothers, as PFAS excretion may occur through milk. As mothers may ingest food and water contaminated with PFASs, it is necessary to consider the dietary exposure from food and water for risk assessment. Notably, PFAS ingestion from drinking water is estimated to contribute 32% of total dietary exposure,[47] a significant proportion resulting from direct consumption or water used during cooking.[48] Accounting for potential exposure from food, the TWI for drinking water may need to be limited to 1.4 ng/kg bw/week.

Both average and worst-case scenario approaches were adopted for risk assessment, benchmarking against TWI and exploring hypothetical exposure levels if consumers were to ingest tap and bottled water. We evaluated the dietary exposure by referring to the dietary consumption data surveyed from the Singapore TDS study conducted in 2021–2023 for 2014 respondents of 15–92 years old.[30] The survey reported that a local respondent consumes 29.448 g/kg bw of tap and drinking water per day. With 98% of household choosing tap water over bottled or barreled water,[49] we assumed the contribution of tap water to be 98% and bottled water to be 2% for the risk assessment. The mean exposure concentration is estimated at 1.74 ng/L, which leads to 0.051 ng/kg bw/day for PFAS exposure through drinking water. Applying the maximum concentration, the exposure concentration is at 6.09 ng/L, leading to 0.18 ng/kg of bw/day of PFAS exposure. Both mean and maximum exposure levels fall below the daily value of 0.2 ng/kg/day (derived from the weekly value of 1.4 ng/kg, Table S14). The results indicate that Singapore consumers face the lowest risk, highlighting the efficacy of Singapore's water quality safeguards as a primary line of defense.

4. CONCLUSIONS

The ubiquitous presence of perfluoroalkyl substances (PFASs) in air, water, and food ecosystems is a global issue. In this study, all of the tap and bottled water samples analyzed met the drinking water quality guidelines established by the EPA and the WHO. Exposure assessment benchmarked against the health-based guidance level set by the EFSA indicated that the exposure risk to PFASs from drinking water for local consumers is low.

Bottled water was found to have the lowest contamination footprint, likely due to stringent processing procedures. The close PFAS contamination level in drinking and mineral water depicts an insignificant difference in PFAS exposure risk when consumers choose between these two types of bottled water. Singapore's tap water consistently remained well below the WHO's safe drinking water quality limits, attributed to the comprehensive water treatment and distribution process island wide.

Given the large volume ever produced and used by the industry and vast diversity of PFAS class of compounds, toxicology evaluation remains limited for many PFASs. The absence of safety and regulatory limits for many PFASs highlights the importance of developing toxicity criteria for groups of PFASs rather than individual compounds. Notably, the prevalent contamination of PFCAs in water urges a thorough understanding of their toxicological profiles.

To advance our understanding and risk assessment capabilities, future studies should consider developing toxicology studies for groups of PFAS analytes based on their chain length. Establishing health-based guidance values for a broader range of PFAS compounds is also critical for a more comprehensive risk assessment. Continual efforts to monitor and assess the potential long-term effects of low-level PFAS exposure through drinking water require attention as well. This approach would provide a more robust framework for assessing and mitigating potential health risks associated with PFAS exposure through drinking water and other sources.

To better safeguard consumer health from PFAS contamination, our future research and studies will focus on the scope expansion of PFAS monitoring and implementation of nontargeted analysis for PFAS in water. It is critical to incorporate emerging PFAS compounds that pose health concerns into monitoring lists, in alignment with evolving toxicological studies. Developing refined classification and monitoring techniques for beverages with complex matrix interference is currently underway.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsestwater.5c00022.

Additional experimental details, materials and methods, targeted PFAS compounds and corresponding internal standards; sampling sources; solvent profile; mass spectrometry; LC retention times and MS parameters; linearity of calibration curves; method detection limits; Kruskal–Wallis (KW) significance test results; Mann–Whitney Wilcoxon test results; global benchmarking; calculated PFAS exposure; sampling point of tap water; importing volume of water by country; chromatogram; and detection frequency (PDF)

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Author Contributions

CRediT: **Ke Tian Tan** data curation, formal analysis, investigation, methodology, validation, visualization, writing - original draft; **Ping Shen** conceptualization, funding acquisition, project administration, resources, supervision, writing - review & editing; **Wei Min Ang** data curation, supervision; **Ignatius Rui Long Lim** methodology, validation; **Wesley Zongrong Yu** data curation; **Yuan Sheng Wu** resources, supervision; **Sheot Harn Chan** resources, supervision.

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Notes

The authors declare no competing financial interest.

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